Both two-dimensional wave-operator problems follow from Xia's uniqueness theorem

Literature-status note for arXiv:2503.23248

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Status: literature_already_answered (direct corollary). Under the hypotheses of Problems 1 and 2, for every nonzero $v = (v_1, v_2) \in \mathbb{R}^2$ the strong limit

$$W_v = \operatorname*{slim}_{t \to +\infty} e^{-it(v_1 B_1 + v_2 B_2)} e^{it(v_1 A_1 + v_2 A_2)}$$

exists, and W_v is independent of v. Hence Problem 1 and Problem 2 both have affirmative answers. This is a direct application of Xia's 2004 wave-operator uniqueness theorem after identifying its non-diagonalizable summand.

The two advertised problems

Let $\alpha = (A_1, A_2)$ and $\beta = (B_1, B_2)$ be commuting pairs of bounded self-adjoint operators on a separable Hilbert space, suppose

$$A_j - B_j \in \mathcal{C}_2^- \qquad (j = 1, 2),$$

and assume the joint spectral measure of α is absolutely continuous with respect to planar Lebesgue measure. Here $\mathcal{C}_2^-$ is the Lorentz $(2, 1)$ ideal. The source asks whether $e^{-itB_1} e^{itA_1}$ has a strong limit as $t \to +\infty$, and whether two directional limits corresponding to distinct angles must coincide [1, Sec. 9, Problems 1–2].

Xia's theorem

For a norm ideal $\mathcal{C}$, Xia decomposes the Hilbert space into a maximal summand on which a commuting self-adjoint tuple is diagonalizable modulo $\mathcal{C}^{(0)}$, the finite-rank closure, and its orthogonal complement. Write $P_{\mathrm{nd}}(\alpha; \mathcal{C})$ for the projection onto the latter summand. A sequence of continuous unimodular functions φ_m is $\mathcal{C}$-admissible for α when

$$\varphi_m(\alpha) P_{\mathrm{nd}}(\alpha; \mathcal{C}) \longrightarrow 0 \quad \text{weakly.}$$

Xia's Theorem 3 says that if $\mathcal{C}$ is any norm ideal other than the trace class, and the coordinate differences of two commuting self-adjoint tuples lie in $\mathcal{C}^{(0)}$, then there is a partial isometry W such that

$$W = \operatorname*{slim}_{m \to \infty} \varphi_m(\beta)^* \varphi_m(\alpha) P_{\mathrm{nd}}(\alpha; \mathcal{C})$$

for *every* $\mathcal{C}$-admissible sequence. In particular the limit is independent of the admissible sequence [2, pp. 145–147, Definition 1 and Theorem 3].

Why the theorem acts on the whole space

Lemma 1. *For the pair α in the source problems,*

$$\mathcal{C}_2^- = (\mathcal{C}_2^-)^{(0)}, \qquad \mathcal{C}_2^- \neq \mathcal{C}_1, \qquad P_{\mathrm{nd}}(\alpha; \mathcal{C}_2^-) = I.$$

Proof. If $s_1(T) \geq s_2(T) \geq \cdots$ are the singular values, then

$$\|T\|_2^- = \sum_{j \geq 1} j^{-1/2} s_j(T).$$

Singular-value truncation makes the tail of this convergent series tend to zero, proving finite-rank density. The diagonal compact operator with $s_j = 1/j$ belongs to $\mathcal{C}_2^-$ but not to $\mathcal{C}_1$, proving the second assertion.

For the last assertion, let $K \neq \{0\}$ be a reducing subspace for α. The restriction $\alpha|_K$ still has absolutely continuous joint spectral measure, with a nonzero multiplicity function m_K. The exact quasicentral-modulus formula recalled in the source gives

$$\left(k_2^-(\alpha|_K)\right)^2 = \gamma_2 \int_{\mathbb{R}^2} m_K(x)\, dx > 0$$

(with $+\infty$ allowed) [1, Sec. 6]. On the other hand, a diagonal tuple has quasicentral modulus zero, and finite-rank density makes the modulus invariant under $\mathcal{C}_2^-$ perturbations. Thus $\alpha|_K$ cannot be diagonalizable modulo $\mathcal{C}_2^-$. No nonzero diagonalizable summand exists, so $P_{\mathrm{nd}} = I$. □

Directional exponentials are admissible

Fix $v \neq 0$ and any sequence $t_m \to +\infty$, and put

$$\varphi_m(x) = e^{it_m v \cdot x}, \qquad x \in \mathbb{R}^2.$$

For vectors ξ, η, the scalar joint spectral measure $\mu_{\xi,\eta}(E) = \langle E_\alpha(E)\xi, \eta\rangle$ is a finite complex measure absolutely continuous with respect to Lebesgue measure. Its Radon–Nikodym density lies in L^1 and has compact support because α is bounded. Therefore the Riemann–Lebesgue lemma gives

$$\langle \varphi_m(\alpha)\xi, \eta\rangle = \int_{\mathbb{R}^2} e^{it_m v \cdot x}\, d\mu_{\xi,\eta}(x) \longrightarrow 0.$$

Since $P_{\mathrm{nd}} = I$, every such sequence is $\mathcal{C}_2^-$-admissible.

Full resolution

Apply Xia's theorem with $\mathcal{C} = \mathcal{C}_2^-$. For every $v \neq 0$ and every sequence $t_m \to +\infty$,

$$e^{-it_m(v_1 B_1 + v_2 B_2)} e^{it_m(v_1 A_1 + v_2 A_2)} \longrightarrow W \quad \text{strongly},$$

where W is independent of the admissible sequence. Sequential convergence for every $t_m \to +\infty$ implies the continuous-time strong limit: otherwise some vector and some sequence of times would violate it. Taking $v = (1, 0)$ proves Problem 1.

For Problem 2 take $v_k = (\cos\theta_k, \sin\theta_k)$, $k = 1, 2$. The preceding argument already shows that both limits exist. To see their equality directly, interlace the two sequences of directional exponentials. The interlaced sequence remains admissible, so Xia's uniqueness forces both subsequences to converge to the same W. This also removes Problem 2's extra assumption that the two limits exist.

Provenance and caution

Xia's theorem predates the source survey by two decades, but the survey does not cite it. The only additional input needed here—the identification $P_{\mathrm{nd}} = I$ for an absolutely continuous pair at the threshold ideal—follows from the exact formula that the survey itself recalls. Thus the mathematical conclusion is a literature corollary rather than a new perturbation theorem. Because the source labels the questions open despite the close match, human review should focus on the $P_{\mathrm{nd}} = I$ identification and on whether the authors intended an unstated stronger notion of wave operator.

References

[1] Dan-Virgil Voiculescu, *Perturbations of operators and non-commutative condensers, an update on the quasicentral modulus*, arXiv:2503.23248v1, 2025.

[2] Jingbo Xia, *On the uniqueness of wave operators associated with non-trace class perturbations*, Canadian Mathematical Bulletin **47** (2004), no. 1, 144–151.